Stony Brook Institute at Anhui University

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Chapter 6
Parameter estimation

Introduction

- In this chapter we will investigate the concept of parameter estimation.
- Our goal is to use sample information to estimate some population parameters.
- The estimation is done in two ways —

  a. point estimation (or single value) and

  b. interval estimation (an interval or range of likely values)

6.1 General concepts of point estimation

- Given a parameter, the purpose of point estimation is to use a sample to calculate a number, which in a sense represents a good guess at the true value of the parameter.
- The resulting numbers are called point estimates.

6.1 General concepts of point estimation

- Let X be a random variable with probability distribution $\{f(x|\theta), \theta \in \Theta)$ which depends on the unknown parameter θ (θ could represent more than one parameter).

6.1 General concepts of point estimation

- Suppose that $\{X_1, X_2, \ldots, X_n\}$ is a random sample of size n from the population X. The value of the estimator $\theta(x_1, \ldots, x_n)$ will be called a point estimate of θ. They can be denoted as $\hat{\theta}$.

Different types of parameters

- Unknown parameters contained in the distribution

For example, $X \sim N(\mu,\sigma^2)$, if μ,σ^2 unknown. The content of parameter estimation is to give their **estimated values** or **value ranges** by constructing statistics

6.1 General concepts of point estimation

◆Function $g(\theta)$ of unknown parameter θ contained in the distribution

For example: $X \sim N(\mu,\sigma^2)$, where μ,σ^2 unknown.
Suppose X is the result of blood test. If we only have interest in the proportion of the people whose test value does not exceed a, that is,

$$P(X \le a) = \Phi\left(\frac{a-\mu}{\sigma}\right)$$

is a function of μ and σ^2.

6.1 General concepts of point estimation

◆Various characteristic parameters of distribution

For example,E(X), Var(X),and other parameters.

The problem of parameter estimation is to estimate the above parameters through estimators using samples obtained from population sampling. Estimators are statistics that estimate population parameters

General concepts of point estimation

- A point estimator T(X) is a statistic intended for estimating a parameter.
- An estimator T(X) is a function of the sample, while an estimate is the realized value of an estimator that is obtained when a sample is actually taken.
- For estimating a parameter θ, usually we denote the estimator as $\hat{\theta}$.

General concepts of point estimation

Point estimation

Suppose that the distribution function of population X is $F(x; \theta)$, $(X_1, X_2, \ldots, X_n)$ is the sample of X, $(x_1, x_2, \ldots, x_n)$ is the corresponding sample value, θ is an unknown parameter in the population distribution.

Question: How to use this sample information to estimate parameters θ?

General concepts of point estimation

Methods: Construct an appropriate statistic $g(X_1, X_2..., X_n)$, and use the observed value $g(x_1, x_2..., x_n)$ of the statistic as the estimate of the true value of the unknown parameter θ.

The point estimator with $g(X_1, X_2..., X_n)$ as θ, is recorded as $\hat{\theta}(X_1, X_2, \cdots, X_n)$
The point estimated value of θ is $g(x_1, x_2, x_n)$, recorded as $\hat{\theta}(x_1, x_2, \cdots, x_n)$

This estimation of the unknown parameter in the population distribution is the point estimation of the parameter.

General concepts of point estimation

Note:
The estimators of the same parameter obtained by different estimation methods may be different.

Questions:

- Which estimator is better for the same parameter?
- What are the criteria for evaluating the estimator?

This section will introduces several common standards: Unbiasedness and Efficiency.

Unbiasedness

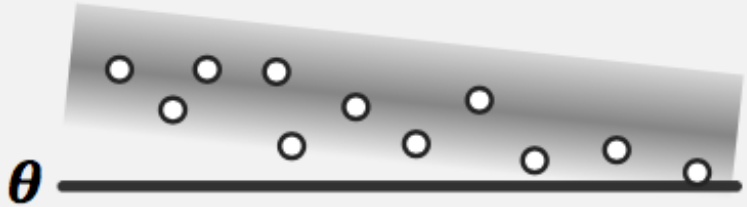
Defintion: An estimator $\hat{\theta}$ is an unbiased estimator of θ

if $E(\hat{\theta})=\theta$ for all θ.

If $\hat{\theta}$ is not unbiased (biased), then $E(\hat{\theta})-\theta$ is called the bias.

unbiased

With bias



$\lim_{n\to+\infty} E(\hat{\theta}_n) = \theta$ asymptotically unbiased

Unbiasedness

Note

- Even if $\widehat{\boldsymbol{\theta}}$ is a unbiased estimator of $\boldsymbol{\theta}$, $\boldsymbol{f}(\widehat{\boldsymbol{\theta}})$ still may not be an unbiased estimator of $\boldsymbol{f}\ (\boldsymbol{\theta})$;
- There may be more than one unbiased estimator for the same unknown parameter.

Unbiasedness

Example 1: Recall Theorem 5.2:

Suppose population X satisfied: $E(X) = \mu$, $Var(X) = \sigma^2$,

so $E(\bar{X}) = \mu, E(S^2) = \sigma^2$.

This theorem means the sample mean and sample variance $\bar{X}$, S^2 are all unbiased estimation of μ, σ^2 respectively.

Unbiasedness

Example 2:

Suppose population X satisfied: E(X)= μ, Var(X) = $\sigma^2 > 0$.

suppose that μ, σ^2 unknown, Show that $\hat{\sigma}_2^{\,2}$ is a unbiased estimator of $\hat{\sigma}^2$ while $\hat{\sigma}_1^{\,2}$ are not.

$$\begin{cases} \hat{\sigma}_1^{\,2} = \dfrac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2 \\ \hat{\sigma}_2^{\,2} = \dfrac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 \end{cases}$$

Unbiasedness

Prove: $\because E(\hat{\sigma}_1^{\,2}) = E\left(\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2\right)$

$$= E\left[\frac{1}{n}\sum_{i=1}^{n}X_i^2 - 2\bar{X}\frac{1}{n}\sum_{i=1}^{n}X_i + \frac{1}{n}\sum_{i=1}^{n}(\bar{X})^2\right]$$

$(\bar{X})^2$

$$= E\left[\frac{1}{n}\sum_{i=1}^{n}X_i^2 - \bar{X}^2\right] = \frac{1}{n}E\left(\sum_{i=1}^{n}X_i^2\right) - E(\bar{X}^2)$$

$$= \frac{1}{n}\sum_{i=1}^{n}\underline{E(X_i^2)} - \underline{E(\bar{X}^2)}$$

Unbiasedness

$$= \frac{1}{n}\sum_{i=1}^{n} \underline{E(X_i^2)} - \underline{E(\bar{X}^2)}$$

$$= \frac{1}{n}\sum_{i=1}^{n} \underline{[D(X_i) + E(X_i)^2]} - \underline{\{D(\bar{X}) + [E(\bar{X})]^2\}}$$

$$= (\sigma^2 + \mu^2) - (\frac{\sigma^2}{n} + \mu^2) = \frac{n-1}{n}\sigma^2 \neq \sigma^2$$

$$\therefore \quad \hat{\sigma}_1^{\,2} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

With bias.

Unbiasedness

While $$E(\hat{\sigma}_2^{\,2}) = E\left(\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2\right)$$

$$= \frac{n}{n-1} E\left[\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2\right]$$

$$= \frac{n}{n-1}\cdot\left(\frac{n-1}{n}\sigma^2\right) = \sigma^2$$

$$\therefore \quad \hat{\sigma}_2^{\,2} = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

$$= s^2$$

Unbiasedness

1. Unbiasedness

Example 3: If population X' s kth moment $\mu_k = E(X^k)$, $k > 0$, then kth sample moment $A_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k$ is the unbiased estimation of μ_k.

which means $E(A_k) = \mu_k$.

1. Unbiasedness

Example 4: If population X~U(0,θ),θ>0. $X_1, X_2, \ldots, X_n$ is a random sample from X.

Show that $2\bar{X}$ and $\frac{n+1}{n} max(X_1, X_2, \ldots, X_n)$ are both unbiased estimators of θ.

Proof:

$$E(2\bar{X})=2E(\bar{X})=2E(X)=2*\frac{\theta}{2}=\theta.$$

so $2\bar{X}$ is an unbiased estimator of θ.

The pdf of $X_{(n)} = \max(X_1, X_2, \cdots, X_n)$ is

$$f(x) = \begin{cases} \dfrac{nx^{n-1}}{\theta^n}, & 0 \le x \le \theta \\ 0, & else \end{cases}$$

so

$$E(X_{(n)}) = \int_0^\theta \frac{nx^n}{\theta^n} dx = \frac{n}{n+1}\theta \qquad \Rightarrow E\left(\frac{n+1}{n} X_{(n)}\right) = \theta,$$

Thus, we prove that $2\overline{X}$ and $\frac{n+1}{n}\max(X_1, X_2, \ldots, X_n)$ are both unbiased estimators of θ.

Example 5: Suppose population X satisfied: E(X)= μ, Var(X) $= \sigma^2$, Show that $\widehat{\mu}_1$ and $\widehat{\mu}_2$ are both unbiased estimators of μ.

$$\widehat{\mu}_1 = \frac{1}{2}X_1 + \frac{1}{2}X_2, \qquad \widehat{\mu}_2 = \frac{1}{4}X_1 + \frac{3}{4}X_2.$$

$$E(\widehat{\mu}_1) = \frac{1}{2}E(X_1) + \frac{1}{2}E(X_2) = \mu.$$

$$E(\widehat{\mu}_2) = \frac{1}{4}E(X_1) + \frac{3}{4}E(X_2) = \mu.$$

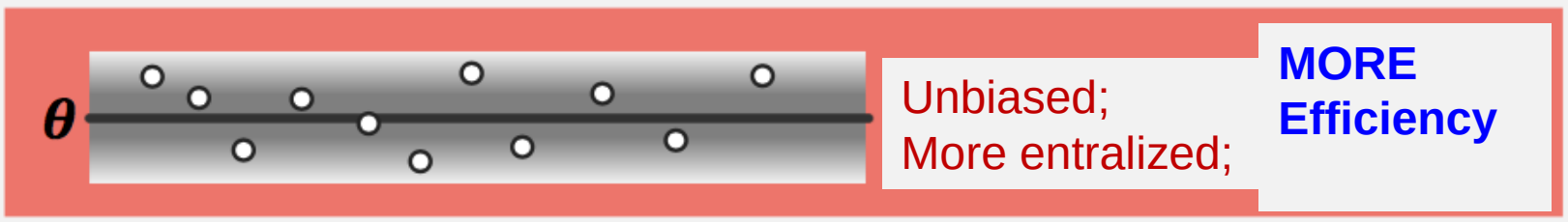
both unbiased

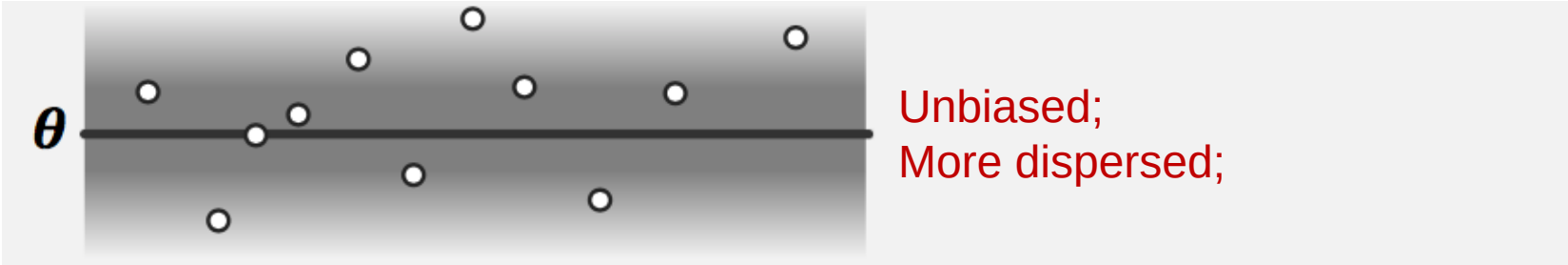
2. Efficiency (or Small variability)

Definition: Let $\widehat{\theta_1}, \widehat{\theta_2}$ be two unbiased estimators for θ with variances, $\widehat{\theta_1}$ is more efficient than $\widehat{\theta_2}$ if

$$\mathrm{Var}(\widehat{\theta_1}) \leq Var(\widehat{\theta_2})$$

This property means the smaller the sampling variability is, the better an estimate will be.





2. Efficiency (or Small variability)

Example 1 : If population X satisfied: $E(X) = \mu, Var(X) = \sigma^2$, then

$$E(X_i)=E(\bar{X})= \mu, \qquad Var(X_i) = \sigma^2, \ Var(\bar{X})=\sigma^2/n,$$

so sample X_i and $\bar{X}$ are both unbiased estimate of μ, but $\bar{X}$ is more efficiency.

2. Efficiency

Exmaple 2: If population X~U(0,θ),θ>0. $X_1, X_2, \ldots, X_n$ is a random sample from X. Show that unbiased estimators $\widehat{\theta_2} = \frac{n+1}{n} max(X_1, X_2, \ldots, X_n)$ is more efficient than $\widehat{\theta_1} = 2\bar{X}$.

Solution:

$$Var(\hat{\theta}_1) = 4Var(\bar{X}) = \frac{4}{n}Var(X) = \frac{\theta^2}{3n},$$

$$Var(\hat{\theta}_2) = Var\left(\frac{n+1}{n}X_{(n)}\right) = \left(\frac{n+1}{n}\right)^2 Var\left(X_{(n)}\right)$$

$$= \left(\frac{n+1}{n}\right)^2 (E(X_{(n)}{}^2) - [E(X_{(n)})]^2) = \left(\frac{n+1}{n}\right)^2 \frac{n}{(n+1)^2(n+2)}\theta^2$$

$$= \frac{1}{n(n+2)}\theta^2,$$

so when n≥2, $Var(\widehat{\theta_1}) > Var(\widehat{\theta_2})$, $\widehat{\theta_2}$ is more efficient.

2. Efficiency

Example 3: Suppose population X satisfied: E(X)= μ, Var(X) $= \sigma^2$, Show that which of $\hat{\mu}_1$ and $\hat{\mu}_2$ is more efficiency.

$$E(\hat{\mu}_1) = \frac{1}{2}E(X_1) + \frac{1}{2}E(X_2) = \mu.$$

$$D(\hat{\mu}_1) = \frac{1}{4}D(X_1) + \frac{1}{4}D(X_2) = \frac{1}{2}\sigma^2.$$

$$E(\hat{\mu}_2) = \frac{1}{4}E(X_1) + \frac{3}{4}E(X_2) = \mu.$$

$$D(\hat{\mu}_2) = \frac{1}{16}D(X_1) + \frac{9}{16}D(X_2) = \frac{5}{8}\sigma^2.$$

$\hat{\mu}_1$ is more efficiency

3. Consistency

Consistency is an asymptotic property of an estimator.

- Definition: An estimator $\hat{\theta}$ *is* is said to be consistent if the probability that $\hat{\theta}$ converges to θ approaches to 1 as the sample size n → ∞, that is

$$\lim_{n\to\infty} P\{|\hat{\theta}_n - \theta| < \varepsilon\} = 1$$

or

$$\lim_{n\to\infty} P\{|\hat{\theta}_n - \theta| \geq \varepsilon\} = 0$$

- Usually moment estimate and MLE have consistency.